



Revealing spatial and temporal patterns of outdoor recreation in the European Alps and their surroundings

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ABSTRACT

Outdoor recreation contributes vitally to human well-being, but spatio-temporal mapping on large scales of this ecosystem service is rarely addressed in a comprehensive manner. In this study, we aim to map recreation supply, demand, and flow, combining different approaches and data sources, including spatially explicit indicators and crowd-sourced information from social media. We analyse spatial and temporal patterns in the European Alps and their surrounding areas (Alpine Space area) and explore societal preferences. Our results indicate that especially mountainous areas provide high ecosystem service supply, while high demand is characteristic of strongly urbanised areas. The spatio-temporal pattern of flow hot spots shows two major trends: recreational landscapes around urban agglomerations are frequented all year round, whereas visitation rates in remote mountain areas depend greatly on the season. By means of a cluster analysis, we identify five types of municipalities, distinguishing municipalities with little importance for recreation, prevailing demand, or supply, and highly used areas. Further, our results suggest that societal preferences can be explained by landscape attributes and tourism infrastructure. In addition to revealing a large-scale spatio-temporal pattern, this study explores methodological possibilities to provide a basis for decision-making and landscape planning regarding recreational ecosystem services.

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1. Introduction

Outdoor recreation in natural and semi-natural environments plays a crucial role for physical and mental health and contributes substantially to human well-being (de Vries et al., 2013; Triguero-Mas et al., 2015). Especially in areas with high urbanisation rates, land degradation, and growing economic wealth, the demand for recreational environments is growing (Guo et al., 2010). Green spaces within or close to urban agglomerations with high population density represent important locations for outdoor recreation activities for many residents (Kabisch, 2015; Langemeyer et al., 2015). Further, mountain regions all over the world are attractive destinations for nature-based recreation and tourism due to their appealing landscapes, access to wilderness and wildlife, and opportunities for outdoor recreation activities like hiking, mountain biking, climbing, or skiing (Pickering and Barros, 2013; Richins & Hull, 2016).

The ecosystem service (ES) 'outdoor recreation' belongs to the group of cultural ecosystem services (CES), which are defined as

the non-material benefits that humans obtain from natural and semi-natural ecosystems (Millennium Ecosystem Assessment, 2005). Although a growing number of studies that address CES focus on recreation and tourism (Hernández-Morcillo et al., 2013; Milcu et al., 2013), a series of challenges related to its assessment remain for two reasons. First, CES depend strongly on human perceptions and subjective preferences that cannot be quantified without analysing societal values (Plieninger et al., 2013). Second, spatially explicit assessment is difficult because CES are not directly linked to ecosystem processes (Burkhard et al., 2012).

Consequently, many studies concentrate on specific environments such as urban areas (Kabisch, 2015; Larondelle & Haase, 2013), agricultural landscapes (Reed et al., 2014; van Berkel & Verborg, 2014), forests (Baerenklau et al., 2010; Verlič et al., 2015), or protected areas (Larsen et al., 2008; Schägner et al., 2016), and fewer studies assess outdoor recreation across different landscapes or at large spatial scales (Helfenstein & Kienast, 2014; Paracchini et al., 2014; Weyland & Laterra, 2014). Hence, assessment approaches often remain limited to the local or regional level (Nahuelhual et al., 2013; Peña et al., 2015; Szücs et al., 2015), which might not reflect the conditions of various landscapes or represent values throughout and across societies, which could

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better provide information for policy and decision-making. On the other hand, large-scale assessments often have to apply simple proxies that fail to account for specific environmental preconditions (Paracchini et al., 2014).

Further, only recently have studies distinguished between ES potential, demand, and actual use (flow) (Burkhard et al., 2014; Villamagna et al., 2013), and CES are less often considered in comprehensive studies due to their subjectivity (Plieninger et al., 2015). Most research focusses in fact on the recreational potential, using survey-based approaches (De Valck et al., 2016; Plieninger et al., 2013) or proxy-based methods (Grêt-Regamey et al., 2015; Helfenstein & Kienast, 2014; Paracchini et al., 2014). Fewer studies have also considered the demand side, using participatory approaches (Beeco et al., 2014; Nahuelhual et al., 2013; Palomo et al., 2013; Zoderer et al., 2016) and economic valuation techniques such as willingness-to-pay (WTP) (Nielsen et al., 2007; Rosenberger et al., 2012) or the travel cost method (Fleming & Cook, 2008). At large spatial scales, mainly empirical or expert-based approaches have been applied to map the demand for outdoor recreation (Paracchini et al., 2014). To assess ES flow in terms of visitation rates, usually costly and time-consuming surveys have been carried out in local studies (Schägnier et al., 2016). At larger scales, novel approaches using social media not only provide promising results for estimating visiting frequencies (Wood et al., 2013), but also identify spatial patterns of landscape values and preferences (Sonter et al., 2016; Tenerelli et al., 2016; van Zanten et al., 2016). However, to facilitate the operationalisation of ES for management, planning, and decision-making, all principal components of the ES chain (supply, demand, and flow) should be addressed (Burkhard & Maes, 2017; Burkhard et al., 2014; Plieninger et al., 2015).

Another issue that has rarely been addressed is that of the temporal dynamics of CES (Hernández-Morcillo et al., 2013). Some examples are related to the dynamics of wildflower blooms (Graves et al., 2016), tourism developments (Guo et al., 2010), seasonality of recreational visits (Wood et al., 2013), impacts of spatio-temporal landscape dynamics on cultural services (Schirpke et al., 2016), and the relationship between the spatial distribution in landscape attributes and temporal changes of visitation rates (Sonter et al., 2016). Nonetheless, spatial and temporal aspects of CES must be considered if trade-offs with other ES and changing environmental conditions or different management scenarios are to be evaluated (Hernández-Morcillo et al., 2013).

In summary, the major challenge of mapping outdoor recreation is related to the spatio-temporal analysis at different scales and extents. Evaluations of spatio-temporal characteristics of 'big data' help to disentangle supply, demand, and flow in the CES assessment. In this study, we use a combination of assessment approaches aiming to map and analyse spatial and temporal patterns of outdoor recreation on a cross-national level. Mapping of ES supply is based on spatially explicit indicators. Local demand considers demographic data, while time-referenced metadata from social media are used as a proxy for ES flow and to explore societal preferences. We evaluate spatial patterns as well as seasonality of outdoor recreation in the European Alps and their surrounding areas (Alpine Space area), and discuss their implications for planning and policy.

2. Materials and methods

2.1. Study area

Our study area corresponds to the Alpine Space Programme cooperation area, including the European Alps and their surround-

ing foothills and lowlands. It covers a surface of about 390,000 km² and comprises Austria, Liechtenstein, Slovenia, and Switzerland as well as several regions of France, Germany, and Italy (Fig. 1, Table S2 in the supplementary material). The Alpine Space area includes 17,042 municipalities of varying size, ranging from 0.006 to 767.7 km² (mean size: 22.77 km²; standard deviation: 30.74) (Appendix Fig. A1). The mountainous core zone is characterised by natural and semi-natural ecosystems (forest, grassland, areas with little or no vegetation), whereas the surrounding lowlands are mainly covered by agricultural land and urbanised areas. In the Alpine Space area live about 70 million people, most of them in great urban agglomerations in the outer belt or peri-Alpine belt (Dematteis, 2009). The Alps are one of the most important global tourist destinations, with about 500 million visitors a year, and tourism is a major source of income from both winter sports and summer tourism (Bartaletti, 2007). The mountainous areas especially offer a variety of opportunities for outdoor recreation, depending strongly on the season, and include activities such as skiing, snowboarding, and cross-country skiing in the winter and hiking, mountain biking, climbing, rafting, and canoeing in the summer.

2.2. Definitions

In this study, we follow the definition of the Common International Classification of Ecosystem Services (CICES) (Haines-Young & Potschin, 2013) on recreation defined as 'physical use of land-/seascapes in different environmental settings', i.e., outdoor activities (e.g., hiking, biking, skiing) for recreational purposes that are practiced in natural and semi-natural ecosystems. According to recent literature (Burkhard & Maes, 2017; Burkhard et al., 2014; Villamagna et al., 2013), we differentiate between the ES supply, demand, and flow:

- Recreation supply includes two aspects: the recreation potential provided by ecosystems and the possibility to benefit from it. Recreation potential is defined as the capacity of ecosystems to provide recreation opportunities due to the natural preconditions without human input and regardless of these being actually used (Burkhard et al., 2014; Villamagna et al., 2013). The service, however, is only provided if people can reach the supply area to carry out recreational activities. Thus, the supply is related to areas providing recreation that are accessible by infrastructure such as roads, trails, and public transport (Ala-Hulkko et al., 2016).
- Demand for recreational opportunities within the society is mainly expressed through stated preferences and values or direct use, but there is no agreement on measures for assessing recreation demand (Wolff et al., 2015). Demand can be mapped considering where beneficiaries live or where the use occurs (Wolff et al., 2015). Here, we map the demand areas within the study area quantifying local beneficiaries (Schirpke et al., 2014). To contribute to the understanding of demand in terms of preferences and values, we also analyse general societal preferences in qualitative terms.
- Flow refers to the actual level of use (Burkhard et al., 2014; Villamagna et al., 2013) and can be measured by the number of people practicing outdoor activities in a defined area and time. It is determined by the population density in the vicinity as well as the capacity of tourism accommodation, as the beneficiaries' origin can be from nearby areas or from other countries or continents. Hence, the benefit of recreational outdoor experiences can be connected to very different spatial scales (Weyland & Lateral, 2014).



Fig. 1. Area covered by the Alpine Space area and borders of included NUTS 2 regions (Nomenclature of territorial units for statistics) (EC, 2015).

2.3. Data collection and mapping methods

2.3.1. Assessing recreation supply

We assessed the supply in three steps: (1) mapping the recreation potential provided by ecosystems, (2) calculating accessibility, and (3) integrating both aspects into one map.

The recreation potential was mapped considering different landscape variables as indicators based on recent literature (Table 1, details in Supplementary material). All data were converted to raster data with a spatial resolution of 100 m. All indicators were considered to equally contribute to the recreation potential and, thus, all indicators were rescaled to 0–100 and overlaid to obtain a recreation potential index (Paracchini et al., 2014). The recreation potential index ranges from 0 (low) to 100 (high).

Accessibility through infrastructure determines whether suitable recreation areas can be used, and proximity to residential areas is a crucial factor for the use of recreational sites (Ala-Hulkko et al., 2016; Kienast et al., 2012; Paracchini et al., 2014; Peña et al., 2015; Weyland & Laterra, 2014). We therefore identified the accessible areas through infrastructure such as paved and unpaved roads, hiking trails, and cycling paths. We further calculated the level of accessibility defined by the proximity to residential areas. Using travel time as a proxy for accessibility, recreation supply was finally mapped by overlaying the recreation potential with the level of accessibility, excluding inaccessible areas. See Supplementary online material for further details.

2.3.2. Mapping recreation demand

We mapped local recreation demand considering the location of beneficiaries of recreational opportunities (Wolff et al., 2015), including both residents and tourists (Schirpke et al., 2014). Since

people have to travel to the supply area to benefit from recreation (Syrrbe & Walz, 2012), proximity is decisive for experiencing recreational services (Ala-Hulkko et al., 2016; Paracchini et al., 2014). Thus, we quantified the local recreation demand based on the number of residents and tourists, using national census data and occupancy rates of tourist accommodation establishments at the municipality level¹. Tourism occupancy rates were converted into a 'permanent resident equivalent' by dividing the number of nights by 365 (Roca, 2016) and added to the number of residents, representing the number of people that might daily benefit from recreational opportunities. Then, we calculated the density per municipality and rescaled the demand from 0 (low) to 100 (high) to compare it with recreation supply and flow.

2.3.3. Quantifying recreation flow

To map ES flow in terms of visitation rates, we relied on meta-data linked to photographs appearing on social media. This crowd-sourced information can be considered a reliable proxy for visiting frequencies and is well-correlated with official visitation statistics (Levin et al., 2017; Sonter et al., 2016; Wood et al., 2013). We used the online photo-sharing website Flickr because it included most pictures related to CES (Oteros-Rozas et al., 2017). Along with the pictures, it provided reliable information to assess recreation flow. Metadata of all geo-tagged public photographs within the boundaries of the study area and uploaded until December 2016 were retrieved by Flickr's public Application Programming Interface (API). We stored the location of the picture, its identification number, the date that it was taken, and the photographer's identifica-

¹ <http://www.statistik.at>, <http://www.bfs.admin.ch>, <http://www.destatis.de>, <http://www.istat.it>, <http://www.insee.fr>, <http://www.stat.si>, <http://www.lv.lv>.

Table 1

Landscape variables used as indicators of outdoor recreation potential.

Landscape variable	Description	Relationship to recreation potential	Data sources	Mapping approach
Naturalness	Index of naturalness (hemeroby)	Preference for more natural environments for outdoor recreation (Peña et al., 2015; Willemen et al., 2008)	CORINE land cover (EEA, 2016a)	Attribution of hemeroby classes to land cover types (Paracchini & Capitani, 2011)
Protected areas	Presence of protected area	Natural environment and high biodiversity (Sonter et al., 2016) and public recreation areas (Paracchini et al., 2014)	Natura 2000 database (EEA, 2015b); Common Database on Designated Areas (CDDA) (EEA, 2015a)	Attribution of scores to IUCN categories (Zulian et al., 2013)
Presence of water	Distance to water bodies	Recreational opportunities (Keeler et al., 2015) and high visual attraction (Arriaza et al., 2004; Ode et al., 2009)	CORINE land cover (EEA, 2016a)	Impedance function of attractiveness (distance <2000 m) from coastlines of sea and lakes (Paracchini et al., 2014)
Landscape composition	Landscape diversity	High recreational and visual attractiveness of diverse landscapes (Kienast et al., 2012; Ode et al., 2009; Schirpke et al., 2016)	CORINE land cover (EEA, 2016a)	Number of land cover types per km ² (Kienast et al., 2012)
Type of relief	Terrain Ruggedness Index (TRI)	Recreational opportunities and visual attraction of rough landscapes (Weyland & Lattera, 2014)	DEM (EEA, 2016b)	TRI classes (Riley et al., 1999)
Mountain peaks	Density of mountain summits	Recreational opportunities (Pomfret, 2011), overview and remoteness (Kienast et al., 2012), and visual attraction (D'Antonio & Monz, 2016)	DEM (EEA, 2016b)	Identification of mountain peaks (Podobnikar, 2012), number of summits per 10 km ²

tion number. Pictures that were located in urban and artificial areas (CORINE land cover classes 111–133) or offshore were excluded from further analysis as we refer only to outdoor recreation in natural and semi-natural environments.

To evaluate the quality of the Flickr data for assessing outdoor recreation, we used a set of 47 municipalities randomly distributed across the study area that contained at least 50 pictures. The related photographs ($n = 10,092$) were examined through a visual content analysis. Photographs taken outdoors and showing recreational facilities, related infrastructures, landscapes or people performing recreational activities were categorised as recreational use. Pictures where the main subject were people not practicing recreational activities, pets, urban infrastructure, motorised vehicles, food, drawings or posters, as well as pure skyscapes or seascapes were classified as non-recreational content. Finally, the ratio between photographs representing outdoor recreation and those showing other contents was calculated for each municipality.

To estimate visitation rates, we first defined so-called ‘user-days’ (Wood et al., 2013), i.e., the number of users who took at least one picture per day in each location. We defined a location by a radius of 1.5 km that refers to the near zone of a specific location (Schirpke et al., 2013) and deleted multiple pictures by the same user and taken the same day, resulting in 1,089,663 pictures. Then, we calculated mean visitation rates per year and season by calculating the point density of all photographs using a moving window with a search radius of 1.5 km (Schirpke et al., 2013) for the respective period. Finally, the recreation flow maps were rescaled from 0 (low) to 100 (high) to overlay them with the recreation supply and demand.

2.3.4. Exploring societal preferences

Human preferences for recreational locations can be linked to landscape attributes such as landscape diversity, trail density or topography through empirical models, using crowd-sourced information or occurrence data (e.g. Braunisch et al., 2011; Sonter et al., 2016; Tenerelli et al., 2016; Westcott & Andrew, 2015). Here again we used the same set of geo-tagged photographs from Flickr to explore general societal preferences and to explain spatial variations in recreation flow.

In a first step, we calculated the distribution of the pictures within CORINE land cover types (EEA, 2016a) and compared the occurrences to the distribution of the land cover types. The second

step included the selection of explanatory variables for the density of pictures within each municipality. We included the six variables used for modelling ES potential and added five context variables (distance from urban areas measured by the travel time from residential areas, number of potential beneficiaries, density of ski lifts, density of hiking and cycling paths, density of tourist attractions such as information points, monuments, castles, viewpoints). All variables were aggregated to the municipality level. Multicollinearity was checked through ordinary least squares (OLS) regression, using variance inflation factor (VIF) and tolerance values. As the residuals of the OLS model were spatially clustered, in a third step, we applied Geographically Weighted Regression (GWR) using the same variables to explain the density of pictures. The six variables related to the potential as well as density of ski lifts and density of other tourism infrastructures were aggregated to three variables because of local multicollinearity. In contrast to global regression models such as OLS, GWR calibrate separate models at each location, considering only the data from the neighbourhood (Fotheringham et al., 2002). Assuming that photo density is context-specific, GWR is suitable to capture spatial variability in the relationships (Tenerelli et al., 2016). The adaptive kernel function was used to define the optimal bandwidth of local relationships. All analyses were performed in ArcGIS 10.4 using the Spatial Statistics extension.

2.4. Statistical analysis

Mean values of recreation supply, demand, and flow were calculated and mapped at the municipality level for all 17,042 municipalities in the study area. We classified all values in three classes (low, medium, and high), using the geometrical interval classification method, to visualise the spatial pattern of each component across landscapes and societies.

To measure the spatial pattern of the three datasets, we calculated spatial autocorrelation using the Moran's I measure (Moran, 1950), which accounts for both locations of municipalities and ES values. This index evaluates whether the spatial pattern is clustered, dispersed, or random. Then, we applied the Getis-Ord G_i^* statistic to identify statistically significant hot and cold spots (Getis & Ord, 1992), i.e., where several municipalities with either high or low values form spatial clusters.

Municipalities with similar patterns of supply, demand, and flow were grouped by carrying out a *k*-means cluster analysis. We first tested the correlations between supply, demand and flow by means of OLS regression models. We then calculated pseudo *F*-statistic values to select the number of clusters representing highest similarities within groups and differences between groups. We used the ‘no spatial constraints’ parameter to group municipalities by data proximity only, disregarding their geographic location and mapped cluster membership. All analyses were performed in ArcGIS 10.4 using the Spatial Statistics extension.

3. Results

3.1. Analysing spatial patterns of outdoor recreation

The number of municipalities with different levels of recreation supply, demand, and flow were distributed unequally in the study area (Fig. 2, Table S3 for NUTS 2 regions and Figs. S1–3 in Supplementary material). Most municipalities (43%) had low supply, whereas it was high in 19% of all municipalities. Demand was medium for 53% and high for only 5% of all municipalities. Medium flow was characteristic of 42% of the municipalities, while low and high flow were equally distributed.

Recreation supply, demand, and flow were spatially rather clustered in different locations within the Alpine Space area (Appendix Table A1), but revealing a diverse spatial pattern (Fig. 3). High supply was concentrated in mountain areas, while low recreation potential was provided by the surrounding lowlands, especially the flat areas. In contrast, high demand was located in urban agglomerations and medium demand in many areas of the peri-Alpine belt. Most Alpine municipalities were characterised by low local demand; only densely populated valleys showed medium to high demand. High flow was concentrated in several mountain locations (e.g., Pennine Alps, Bernese Highlands, Bavarian Alps, Karwendel, Dolomites), in proximity to great lakes (e.g., Lake Geneva, Lake Como, Lake Garda) and urban agglomerations (e.g., Zurich, Milano, Munich, Vienna), and along the coastline of the Mediterranean Sea in France and Italy (Liguria). Flow decreased with increasing distance from areas with high flow; low flow was located mainly in the western (French lowlands) and eastern (Niederösterreich, Steiermark, Zahodna Slovenija) part of the Alpine Space area.

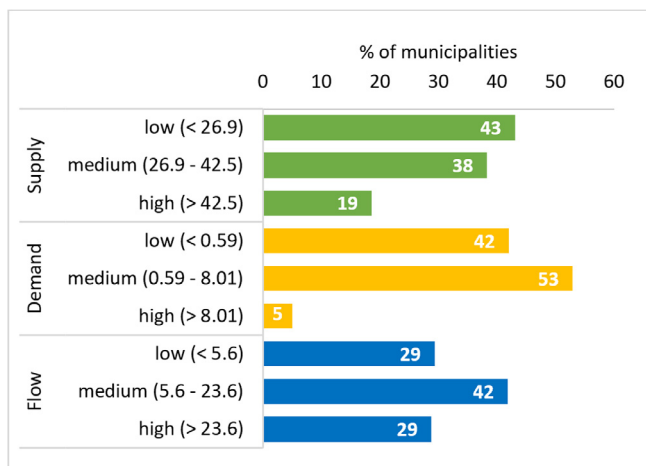


Fig. 2. Distribution of recreation values within the municipalities of the Alpine Space area. Thresholds were obtained from the geometrical interval classification method.

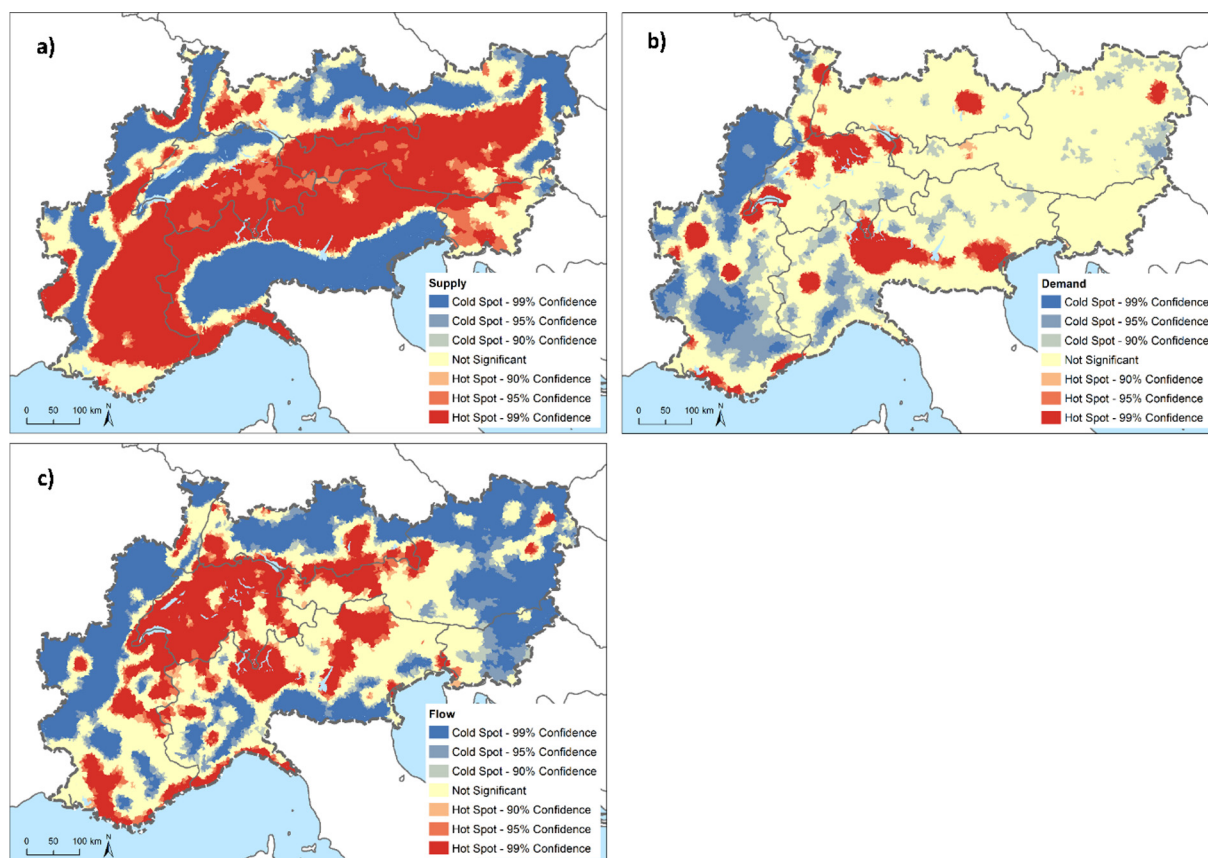


Fig. 3. Hot and cold spots of (a) supply, (b) demand, and (c) flow maps at the municipality level for the Alpine Space area.

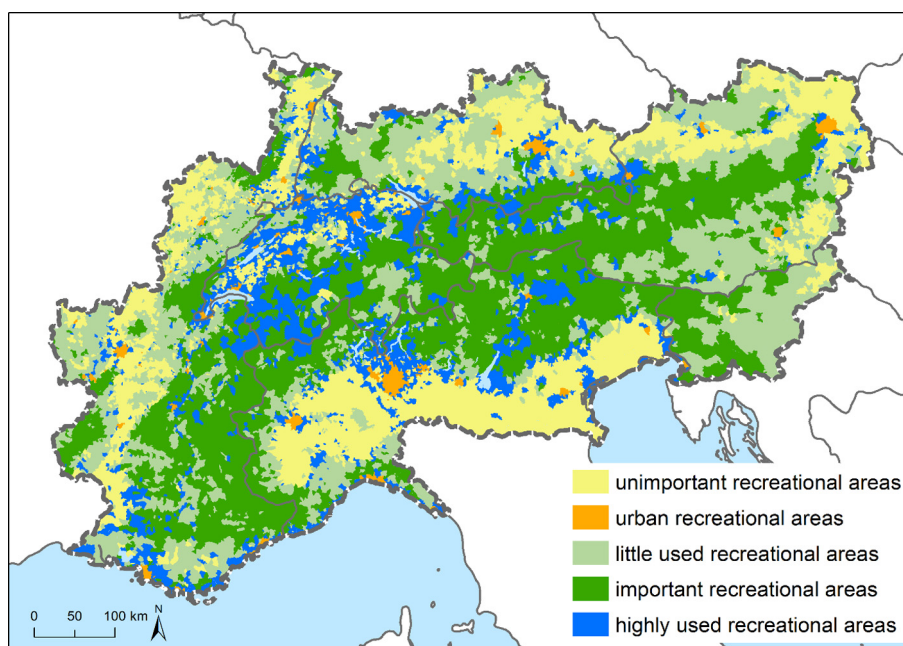


Fig. 4. Municipalities with a similar pattern of supply, demand, and flow, produced by cluster analysis.

Table 2

Statistics for each component for the five clusters.

			Mean	Std. Dev.	Min	Max
Unimportant recreational areas	N = 5059	Supply	16.98	5.11	0.00	28.68
		Demand	1.59	1.88	0.02	14.87
		Flow	8.67	7.14	0.00	32.20
Urban recreational areas	N = 336	Supply	23.27	11.78	0.00	70.59
		Demand	22.92	12.55	12.52	100.00
		Flow	38.67	15.30	0.00	81.73
Little used recreational areas	N = 4851	Supply	32.47	5.27	23.98	46.45
		Demand	0.84	1.07	0.00	14.85
		Flow	7.97	5.89	1.00	24.42
Important recreational areas	N = 3411	Supply	48.16	6.63	34.56	78.24
		Demand	0.55	0.80	0.00	12.26
		Flow	18.86	10.10	1.00	53.01
Highly used recreational areas	N = 3355	Supply	31.16	10.16	7.43	74.70
		Demand	3.21	3.06	0.00	14.80
		Flow	37.93	9.39	16.49	81.49

The OLS results demonstrate that the recreation flow was rather independent from supply and demand ($R^2 = 0.05$ for the supply, $R^2 = 0.12$ for the demand, and $R^2 = 0.20$ for the combination of supply and demand), although the relationship was positive in all cases. Thus, we identified five groups of municipalities, differing in the levels of supply, demand, and flow (Fig. 4, Table 2). The 'unimportant recreational areas' cluster, characterised by generally low ES values, were located in rural areas of the peri-Alpine belt, and 30% of all municipalities belonged to this group. The cluster 'urban recreational areas' (2% of the municipalities) matched densely populated areas with high demand and flow, but medium supply. In contrast, 28% of all municipalities, mainly located in less frequented mountain areas, were part of the 'little used recreational areas' cluster, representing medium to high supply, but low demand and flow. The 'important recreational areas' cluster indicated high supply with medium flow but low demand; it included 20% of the municipalities, situated in the core zones of mountain areas. 'Highly used recreational areas' corresponded to 20% of the municipalities, indicating medium to high supply and high use. These municipalities were often located in proxim-

ity to urban agglomerations, but also close to high-supply areas such as large lakes, famous mountain ranges, or the Mediterranean Sea.

3.2. Identifying spatio-temporal dynamics of recreation flow

We assessed the seasonality of visitation rates, distinguishing between spring (March to May), summer (June to August), autumn (September to November), and winter (December to February) (Figs. S4–7 in Supplementary material). Fig. 5 demonstrates that the seasonal pattern of uploaded Flickr data was consistent over time. To these data, monthly totals were summed and decomposed into the underlying patterns of trend, seasonality, and noise, following a multiplicative scheme. On average, recreation flow was highest in summer (36%), whereas lowest visitation rates occurred in winter (18%). Spring and autumn had similar flow (about 23%).

The spatial pattern and autocorrelation of ES flow varied for each season (Fig. 6, Appendix Table A1). In summer, high visitation rates prevailed over large part of the Alps, especially in the western

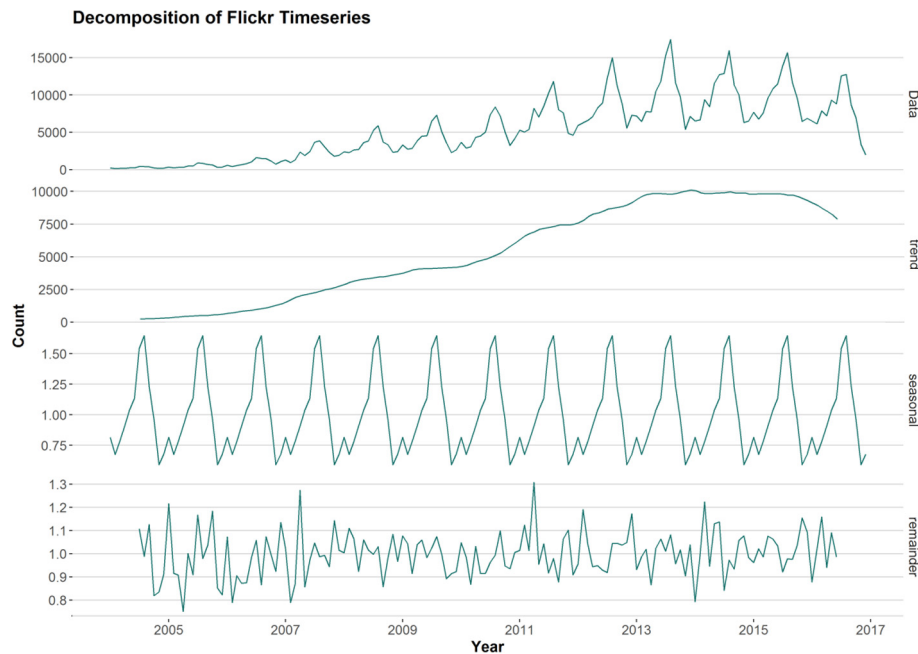


Fig. 5. Seasonal pattern in time series of Flickr data.

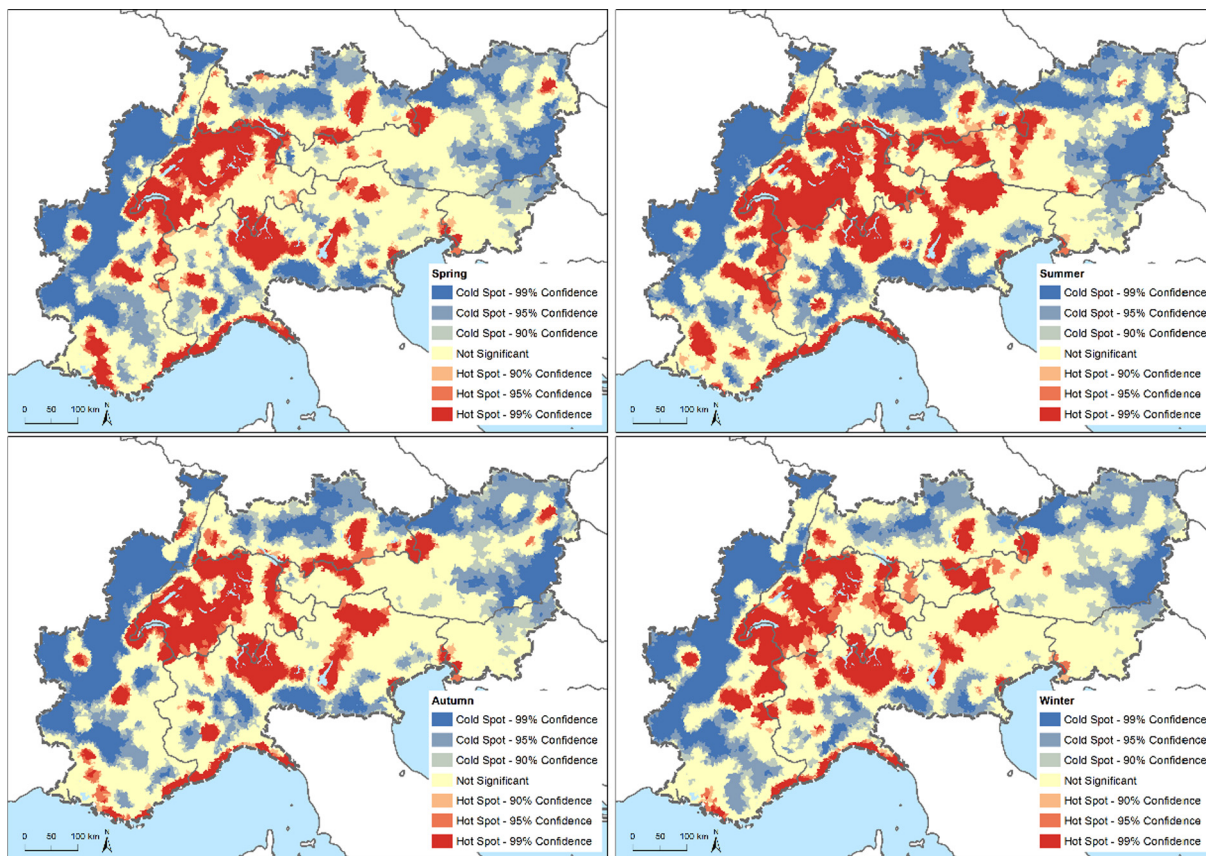


Fig. 6. Seasonal pattern of hot and cold spots of recreation flow in the Alpine Space area.

and central Alps. In contrast, clustering was lower in winter, and ES flow concentrated on rather isolated spots, corresponding to important ski resorts. Especially hot spots around urban agglomerations (e.g., Lake Geneva, Milano, Zurich) were persistent over time, whereas hot spots located for example in remote mountain

areas (e.g., Pennine Alps, Dolomites) depended on the season and were least expanded in spring.

The visual content analysis for the sample municipalities indicated that on average 80% of the photographs were related to outdoor recreation across the study site (Appendix Table A2, Fig. 7).

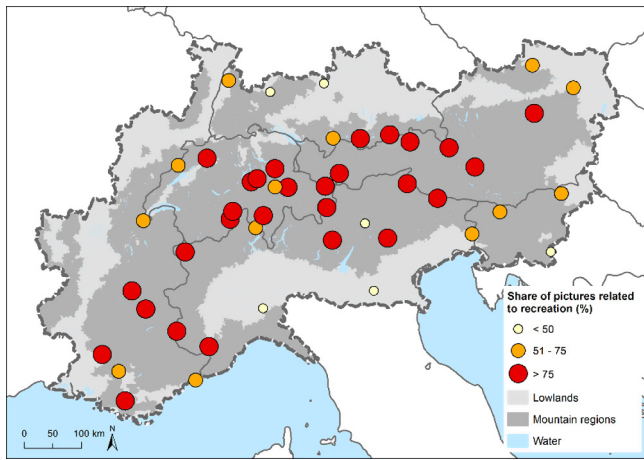


Fig. 7. Location of sample municipalities and results of the visual content analysis of the related Flickr photos.

The share on municipality level varied between 28% and 97% (mean 75%, standard deviation 21%) and was generally higher in mountainous areas than in lowland regions.

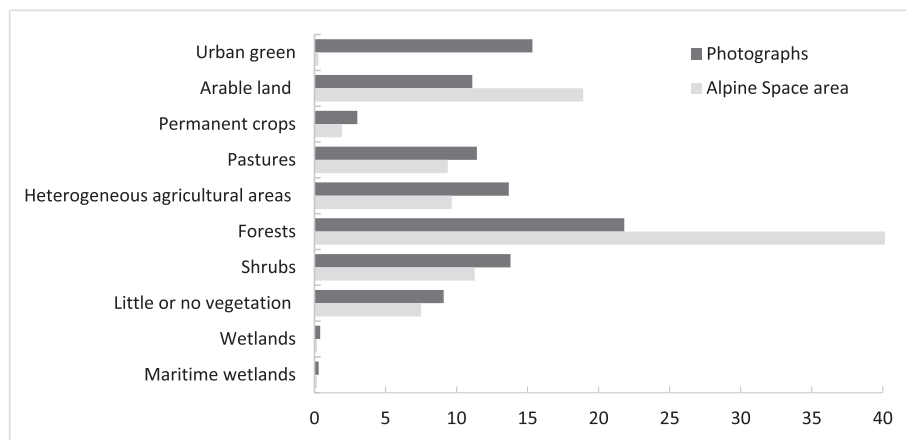


Fig. 8. Distribution of photographs from Flickr in relation to different land cover types (CORINE land cover, 2nd level).

Table 3
Results of the OLS regression analysis, including collinearity diagnostics.

Variable	Coefficient	Std Error	t-Value	Probability	Robust SE	Robust t	Robust Pr	VIF
Intercept	15.4244	0.5448	28.3140	0.0000 [*]	0.5764	26.7592	0.0000 [*]	–
Naturalness	–0.2453	0.0099	–24.7064	0.0000 [*]	0.0114	–21.4597	0.0000 [*]	4.4315
Land use diversity	0.0320	0.0153	2.0901	0.0366 [*]	0.0161	1.9843	0.0472 [*]	1.1088
Peak density	–0.1019	0.0150	–6.7844	0.0000 [*]	0.0138	–7.3634	0.0000 [*]	3.0526
Terrain ruggedness	0.1591	0.0073	21.9295	0.0000 [*]	0.0080	19.9647	0.0000 [*]	5.0555
Protected areas	0.0168	0.0029	5.8390	0.0000 [*]	0.0029	5.7367	0.0000 [*]	1.1754
Distance to water	0.1648	0.0051	32.1857	0.0000 [*]	0.0071	23.3776	0.0000 [*]	1.0592
Ski lifts	56.5492	2.1761	25.9862	0.0000 [*]	1.9580	28.8806	0.0000 [*]	1.2365
Hiking and cycling paths	10.4474	0.2426	43.0694	0.0000 [*]	0.2923	35.7455	0.0000 [*]	2.0405
Potential beneficiaries	0.2978	0.0263	11.3063	0.0000 [*]	0.0460	6.4699	0.0000 [*]	1.8015
Distance from urban areas	–0.0745	0.0196	–3.7970	0.0002 [*]	0.0173	–4.3117	0.0000 [*]	1.8441
Tourist attractions	2.0616	0.1305	15.7929	0.0000 [*]	0.1611	12.7974	0.0000 [*]	1.6261
Koenker (BP) Statistic	1187.89			Prob (>chi-squared), (11) degrees of freedom			0.0000 [*]	
Jarque-Bera Statistic	2603.46			Prob (>chi-squared), (2) degrees of freedom			0.0000 [*]	
Number of Observations	17042			Akaike's Information Criterion (AICc)			129,735.09	
Multiple R ²	0.43			Adjusted R ²			0.43	
Joint F-Statistic	1181.27			Prob (>F), (11,17030) degrees of freedom			0.0000 [*]	
Joint Wald Statistic	9495.16			Prob (>chi-squared), (11) degrees of freedom			0.0000 [*]	

^{*} = $p < 0.05$.

3.3. Explaining societal preferences

Numbers of posted photographs reflected an uneven distribution across different land cover types (Fig. 8). People took most photographs (88%) in open landscapes, of which 39% were in agriculturally used land, 24% in rather natural land cover types, and 15% in green urban areas. The number of pictures in green urban areas exceeded the proportional share of this land cover type by many times (15% vs. 0.3%), whereas only 22% of all pictures were taken in forests (40% of total area).

Eleven variables explained 43% of the variation in photo density ($R^2 = 0.43$, adjusted $R^2 = 0.43$). Model coefficients suggest that photo density was positively related to high density of hiking and cycling paths, ski areas, proximity to water, terrain ruggedness, whereas naturalness and peak density affected negatively photo density ($p < 0.001$) (Table 3). The OLS regression residuals showed strong spatial autocorrelation (Moran's I : 0.55, z -score: 27.42, p -value: < 0.001), therefore, a GWR model was applied to describe local relationships. The GWR model improved model performance over the OLS model: the adjusted R^2 increased from 0.43 to 0.73 and the AIC value decreased from 129,735 to 48,848 (Tables 3 and 4). Having a lower AIC value with a difference greater than 3, the GWR model describes better the relationships than the OLS

Table 4
Results of the GWR model.

Variable	Mean	Minimum	Maximum	Standard deviation
Intercept	11.9376	−130.2936	121.7602	21.6014
Potential 1 ¹	0.2738	−2.6514	5.6242	0.6068
Potential 2 ²	−0.2155	−4.4450	2.9838	0.5996
Path density	7.3182	−181.0778	181.4398	17.2504
Tourist attractions	16.7045	−423.2657	395.0862	37.1436
Distance from urban areas	−0.1197	−23.7860	19.3672	2.7516
Potential beneficiaries	0.7143	−31.7793	59.3366	3.3432
Number of neighbours			75.00	
Residual Squares			334,368.00	
Effective Number			1628.70	
Sigma			8.05	
Akaike's Information Criterion (AICc)			48,848.48	
R ²			0.80	
Adjusted R ²			0.73	

¹ Terrain ruggedness, land use diversity, protected areas, and distance to water.

² Naturalness and peak density.

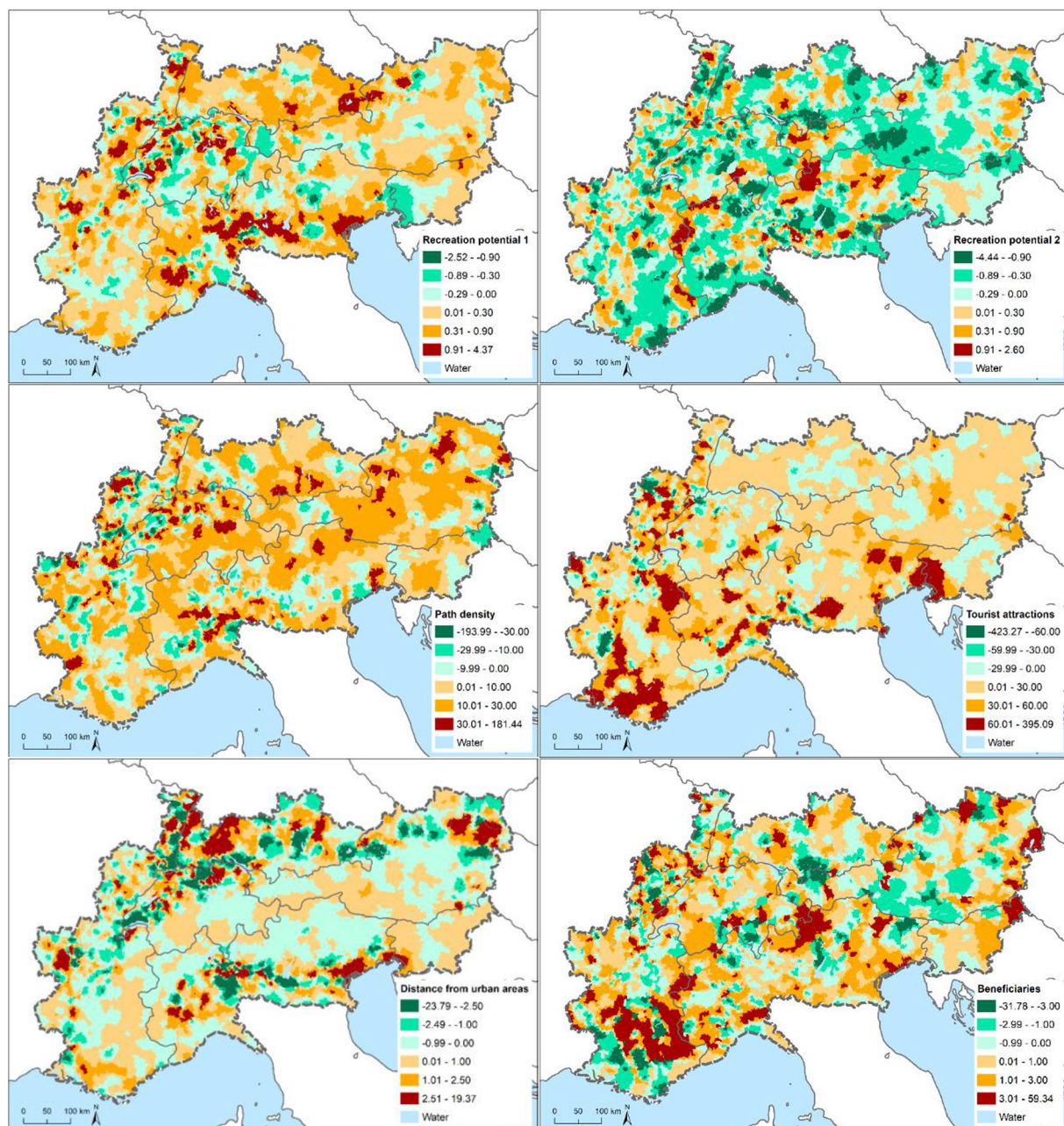


Fig. 9. Spatial variation in the estimated regression coefficients of the variables included in the GWR model. Recreation potential 1 includes terrain ruggedness, land use diversity, protected areas, and distance to water, whereas recreation potential 2 comprises naturalness and peak density.

model (Fotheringham et al., 2002). The results of the global Moran's I test of the residuals for spatial autocorrelation indicated a random distribution (Moran's I: 0.07; z-score: 3.34; p -value: <0.001), and spatial dependencies in the OLS model have been removed with GWR.

The spatial distribution of the local coefficients of each variable of the GWR revealed that the influence of most variables varied across the study site to different degrees (Fig. 9), varying significantly for the five clusters, especially between urban recreational areas and little used and important recreational areas (Appendix Fig. A2; Table S3 and Figs. S8–12 in the Supplementary material). Whereas recreation potential 1 (terrain ruggedness, land use diversity, protected areas, and distance to water) was positively related to recreational use mainly in lower areas in Switzerland, Germany and Italy, recreation potential 2 (naturalness and peak density) showed rather negative influence across the study area, except for several less accessible areas. A positive influence of path density prevailed over large areas, while tourist attractions had strong positive influence mostly in France and Italy. Distance from urban areas was important only in the surrounding lowlands, where large urban areas were located, demonstrating great local differences. Spatial pattern of the influence of beneficiaries were similar to recreation potential 2, but relationships were more positive.

4. Discussion

Assessment of CES remains challenging due to their dependency on human perceptions and preferences (Plieninger et al., 2013) and their disconnectedness to ecosystem processes (Burkhard et al., 2012). In this study, we addressed challenges related to the mapping of outdoor recreation, analysing spatial and temporal patterns of supply, demand, and flow across different landscapes and societies.

4.1. Spatial and temporal pattern of outdoor recreation

Our results revealed great differences in spatial pattern between supply, demand, and flow. Whereas outdoor recreation is provided mainly by mountain areas (Oteros-Rozas et al., 2017; Peña et al., 2015; Egarter Vigl et al., 2017), major demand areas (beneficiaries) correspond to large urban areas in the surrounding lowlands (Syrbe & Walz, 2012; Wolff et al., 2015). Hot spots of recreation flow showed different temporal patterns depending on the geographical position. Recreational areas close to urban agglomerations, especially large lakes and mountain areas, were frequented all year round, indicating that primarily residents use them (Kabisch, 2015; Larondelle & Haase, 2013). In contrast, flow hot spots in remote mountain areas demonstrate high seasonal variability, underlining the particular challenging nexus of outdoor recreation, tourism and climate change in mountain tourist destinations (Bonzanigo et al., 2016).

In combination, the results of the cluster analysis and the GWR models provide suitable information for management and planning, even without a clear understanding of the drivers. Landscapes in proximity to urban areas that have high recreational use are increasingly under pressure due to increasing population and land transformations (Koppen et al., 2014). To ensure access to recreational areas for everyday life, areas with a negative relationship between supply and use rates might offer alternatives, as these areas offer high natural potential but may lack the necessary infrastructure. Thus, management actions influencing visitor flows depend on the balance between recreation potential and accessibility in terms of infrastructure. In mountain areas, municipalities belonging to 'little used recreational areas' may enhance socio-economic well-being by promoting the recreational values

provided by the natural environment and by creating the needed tourism infrastructure such as accommodations, restaurants, hiking paths, and signs (Schirpke et al., submitted). However, as the GWR results indicate, in areas with a negative relationship between recreational use and path density the development of the path network will not improve the recreational service. Furthermore, overbuilding (Needham et al., 2011) or a large number of visitors (noisiness, crowdedness) negatively affect recreational quality (Scolozzi et al., 2014). Increasing visitor numbers, in particular in areas with a negative relationship between naturalness (recreation potential 2) and flow, may also amplify negative impacts on the environment, leading for example to higher disturbance of wild animals or degradation of vegetation and soil (Peksa & Ciach, 2015; Rixen et al., 2013; Steven et al., 2011). Even more severe effects can be expected for 'highly used recreational areas', matching hot spots of recreation flow close to urban agglomerations, large lakes or the Mediterranean Sea, and famous mountain ranges, which demonstrate negative relationships to naturalness. Hence, an integrated management at the regional level, controlling the strength and direction of the relationship between supply and flow, can regulate the impacts of recreational use on the environment by applying management interventions such as creating more sustainable recreation sites, managing the spatial and temporal distribution of visitors, and raising awareness for activities with lower impacts (Marion, 2016).

In general, the regression results suggest that recreational flow is neither strongly linked to supply nor to demand in our study area. Although natural assets are of great importance, the actual use of recreational areas is highly influenced by human inputs (e.g. touristic infrastructure, promotion of destinations) and perceptions (e.g. degree of popularity) (Hallmann et al., 2014; Schägner et al., 2016). The recreational flow resulted to be also rather independent from demand areas, except for green urban areas. The demand was generally positively associated to recreational use, but the GWR analysis revealed great variations, especially in high-elevated municipalities, which can be linked to reduced accessibility. When combining supply and demand, the explanatory power was higher than for each single component, but still not sufficient to accurately predict patterns of flow over the study area.

4.2. Using social media to assess societal preferences

To integrate the quantitative mapping of recreation demand, we used photo density based on social media also to assess general societal preferences in qualitative terms. The variations of the distribution of photographs across different land cover types can be related to several factors. The high share of photographs in green urban areas may be influenced by proximity to residential areas and good accessibility (Kabisch, 2015; Langemeyer et al., 2015). The lower photo density in forest and agricultural land may be result from lower scenic values (Schirpke et al., 2016). Nevertheless, forest areas are highly appreciated areas for recreational activities (Baerenklau et al., 2010; Edwards et al., 2011), and photo density might not reflect good enough their recreational values.

Although the OLS model had low explanatory power ($R^2 = 0.43$), similar to other studies (Keeler et al., 2015; Sonter et al., 2016; Tenerelli et al., 2016; Wood et al., 2013), we found significant relationships between photo density and landscape features as well as context variables revealing general societal preferences. In line with other studies (Keeler et al., 2015; Oteros-Rozas et al., 2017), our results suggest high preferences for water in the landscape. Especially great water bodies offer a wide range of recreational activities (Keeler et al., 2015). Terrain ruggedness increases recreational use because of more opportunities and high visual attraction (Kienast et al., 2012; Pomfret, 2011; Weyland & Laterra,

2014). The OLS model also confirmed that protected areas are important recreational areas which can be explained by their natural environment, high biodiversity (Arnberger et al., 2012; Siikamäki et al., 2015; Sonter et al., 2016), and public recreation opportunities (Paracchini et al., 2014). People prefer natural mountain landscapes with forests and water bodies for recreational activities such as wildlife observation, hiking and trekking (Bell et al., 2007; Pastorella et al., 2017). The GWR analysis revealed strong regional variation of ES potential across the entire study area, suggesting that people are highly susceptible in the selection of destinations for outdoor recreation through destination management, as visitor satisfaction depends on greatly on the services offered, accessibility, hospitality, and the image of the destination (Hallmann et al., 2014). In fact, we found a strong positive relationship between photo density and hiking and cycling paths, which emphasises the importance of touristic infrastructure for recreational activities (Sonter et al., 2016). In addition, the presence of ski lifts in mountain areas increases recreational activities such as hiking and skiing (Oteros-Rozas et al., 2017; Pickering and Barros, 2013). In contrast to other studies (Peña et al., 2015; Sonter et al., 2016; Willemen et al., 2008), we found a negative relationship between photo density and naturalness, which can be explained by lower accessibility to large areas characterised by high elevation and low human influence in our study area. For the same reason, people took generally less pictures in areas with high peak density, whereas terrain ruggedness had a positive effect on photo density. As the recreational use is determined to a certain degree by the location of demand areas (Kienast et al., 2012), we found a positive influence of close urban areas on photo density.

4.3. Methodological insights

To model the potential supply, we included six key variables that we considered decisive for our study area, whereas other large-scale assessments used fewer (Paracchini et al., 2014) or even more variables (Weyland & Latererra, 2014). The OLS confirmed the influence of the indicators that were used for assessing recreation potential. The negative relationship between flow and naturalness as well as peak density underlines the role of accessibility for the actual use (Ala-Hulkko et al., 2016), and can help to identify potential areas for outdoor recreation. On smaller scales, other variables might be more important, and information that is more detailed may produce more meaningful maps on the local level; for example, CORINE land cover might be enhanced with data on forestry and agricultural use (Rüdiger et al., 2012). For specific applications, it might also be of interest to distinguish between different types of activities, e.g., hiking requires other terrain than mountaineering; winter sports depend on snow cover (Gilaberte-Búrdo et al., 2014), while summer activities are little influenced by climatic conditions (Pröbstl-Haider et al., 2015). To evaluate the supply, we integrated data on accessibility to potential recreational areas (Ala-Hulkko et al., 2016). We analysed travel time (by car and foot) as a proxy for accessibility. Real travel time by car, however, depends on road conditions, weather, and traffic, while walking speed varies with physical fitness, slope, path, or terrain conditions (Irmischer & Clarke, 2017).

The local recreation demand does not account for the physical, social and economic heterogeneity of the society and hence potential differences in behaviour, as we assumed that each person had the same possibility to carry out recreational activities. Differences in frequency of activities may occur between social groups, as well as between the local population and tourists. Thus, we might have overestimated the local demand of the residents, but we could account for demand outside the area represented by tourists (Wolff et al., 2015). In a sensitivity analysis, we tested the variations in demand under the assumption that residents on average

practice recreational activities two days per week, whereas tourists have the possibility for outdoor recreation seven days per week. On the large scale of our study area, this aspect did not influence spatial pattern; on average, this reduced recreational demand by 0.2%. However, studies on smaller scales should evaluate differences in the behaviour of residents and tourists to accurately assess ES demand. Furthermore, sport participation behaviour varies across European countries, declining when going from north to south and from west to east, and is influenced by social characteristics such as gender, age, income, education (Van Tuyckom and Scheerder, 2010). Ideally, residents should therefore be stratified based on differences in recreation behaviour, and tourists that practice recreational activities should be separated from other overnight guests.

Since our results indicate that the local demand is a rather weak driver for recreational flow, it is questionable how to quantify demand, which might also depend on the local context. In mountain areas that are dominated by the use of tourists, the demand might be estimated more accurately by mapping the level of use, because people usually stay close to the selected destination, although not necessarily in the same municipality. In contrast, the demand in urban areas might be better reflected by mapping the beneficiaries, as people may use urban parks for every-day recreational activities and the surroundings for day-trips.

To approximate ES flow, we used crowd-sourced information from Flickr, which hosts a large repository of photographs that represent recreational activities (Oteros-Rozas et al., 2017). The representativeness of social media has been critically discussed, as social media use is associated with specific groups of people rather than the population as a whole (Oteros-Rozas et al., 2017; Pastur et al., 2016; Tenerelli et al., 2016; Wood et al., 2013). Frequent use of digital technologies, higher income, and higher levels of education are characteristics of people who share georeferenced photographs and typically represent urban rather than rural recreationists (Li et al., 2013). Preferences and visitation rates may be biased by the type of recreational activities or by the preferences of local recreationists vis-à-vis those of tourists (Wood et al., 2013). By integrating other social media platforms, bias in the Flickr data might be reduced, even though different platforms revealed similar patterns (Van Zanten et al., 2016). Furthermore, the underlying geotagging mechanisms and unidentified uploading behaviour represent uncertainties within the spatial distribution of the meta-data. Consequently, further research should consider applying an image classification scheme, allowing a more precise evaluation of the respective recreation activity.

Nevertheless, we argue that not only the content of the photo is important, but also the fact that people take pictures in certain places. People document their recreational trips by taking pictures of scenic views, landscape features (e.g. waterfalls, glaciers), touristic infrastructures (e.g. trail network, ski lifts, path signs), plants, animals, or themselves while they hike, cycle, ski or carry out other activities. Indeed, Flickr photos have been demonstrated to be highly correlated with visitation statistics (Levin et al., 2017; Sessions et al., 2016; Sonter et al., 2016; Wood et al., 2013). We estimated ES flow using relative values on a large spatial scale, as we intended to analyse spatial patterns of recreational use and the relationships with supply and demand rather than but to assess real visitation rates, which should be integrated and validated with observed visitor numbers (Wood et al., 2013). Due to the large extent of our study area and the high number of photographs, we could not select visually the photographs representing recreational activities and filtering the pictures based on tags led to too less pictures because of many missing tags or unclear classification. Hence, we had to include also pictures that did not refer to outdoor recreation, but the results from the sample municipalities suggested that a very high share of photographs (80%)

were related to outdoor recreation across the study site, especially in mountainous areas (Appendix Table A2, Fig. 7). In municipalities located in the surrounding lowlands of the European Alps or in more urbanised municipalities, the share of pictures related to outdoor recreation seems to be tendentially lower as in mountainous and rural municipalities. This may have led even to an overestimation of recreational use in these areas, but the different behaviour of local people in taking and uploading pictures to social media may compensate this effect. High photo density may also result from the presence of monuments of historical, cultural or religious value, such as castles and churches, or special infrastructures like scenic train routes or spectacular pass roads, attracting many visitors. Although visits to cultural attractions are not included in the ES outdoor recreation in sensu stricto, these constructions are embedded into the landscape and part of outdoor tourism, if not located in urban areas.

Aesthetic landscape values influences the number of photographs taken in a specific location (Figuerola-Alfaro & Tang, 2017), as people may take more photos in scenic places. Recreational and aesthetic values are often considered as overlapping and assessed together (Van Zanten et al., 2016). Consequently, less scenic locations might be underrepresented when focusing on recreational activities, and hence, the level of recreational use may be underestimated in these areas. On the other hand, a beautiful landscape is an important factor for selecting the destinations for nature-based recreation and tourism (Scolozzi et al., 2014). Bias in the data was reduced by calculating user-days, which eliminated also multiple pictures by the same user taken in the same location, and by aggregating the data to the municipality level.

In summary, geo-tagged photographs of social media are useful to estimate recreation flow over large areas, at least on a relative scale, but these data reflect better touristic use than everyday recreational activities and should therefore be carefully used in less touristic areas. Despite the above-discussed limitations, the Flickr data can further provide insights into societal preferences for large study areas by linking them to landscape attributes. Future research should analyse the content of a picture sample and possibly be integrated with specific surveys for deeper understanding of human perceptions and preferences, for which this study can serve as a valuable information basis. On small spatial scales, pictures can be analysed visually, but further efforts are needed to automatically classify picture contents on large scales.

5. Conclusions

Our analysis shows that through a combination of assessment and mapping approaches, recreation supply, demand, and flow can be accurately represented across different landscapes and societies. Crowd-sourced information, especially, was shown to have a high potential to assess CES flow over large areas, with lower costs than for empirical studies. It is shown that by focussing solely on the metadata of freely available picture datasets, enough information can be extracted to accurately assess the use of recreational ecosystem services over large areas. Further efforts should be undertaken to analyse societal preferences by classifying picture contents.

ES mapping at the landscape level has revealed a spatio-temporal pattern of CES, indicating spatial relationships between supply and demand areas. Here, we provide a modern assessment scheme for data-sparse regions, potentially providing important

insights for the processes of decision-making and landscape planning. In mountain areas, recreation supply is generally at high levels, but the actual use differs greatly between locations. Nonetheless, consistent datasets at low resolution in a transnational context about the exploitation coefficients of recreational supply are nonexistent. However, little-frequented areas might benefit from this knowledge. Efforts to increase visitor numbers in unpopular areas can be fostered, whereas highly frequented locations can profit of an adequate visitor management to maintain recreational quality and to avoid negative impacts on the environment. In the face of global climate change, rethinking the tourism concept might be necessary to support socio-economic development. Here, municipalities and regions, relying on tourism, potentially benefit from increased knowledge and accurate prediction of the seasonal effects of recreation flow. In contrast, recreational landscapes in proximity to urbanised areas are year-round destinations for the local population. This information should not be neglected in landscape planning processes.

To define and implement strategies for managing recreation and other ES, the results of this study provide a valuable basis for further research at the local level, considering specific environmental conditions and the socio-economic context. Future studies may also distinguish between different recreational activities and should address future developments, such as increasing demand on the one side and landscape and climate change on the other side.

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Appendix A.

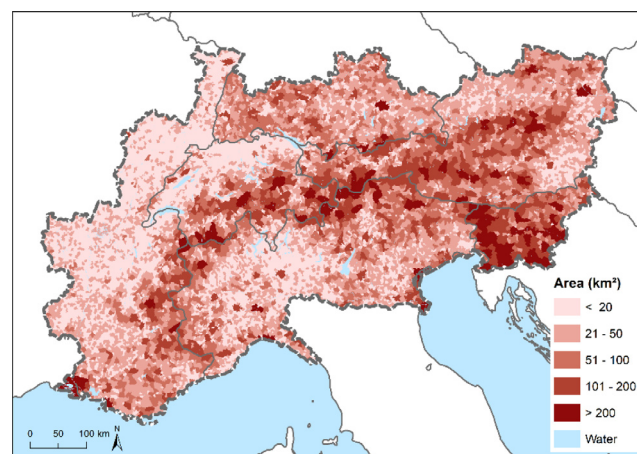


Fig. A1. Size of municipalities in the Alpine Space area.

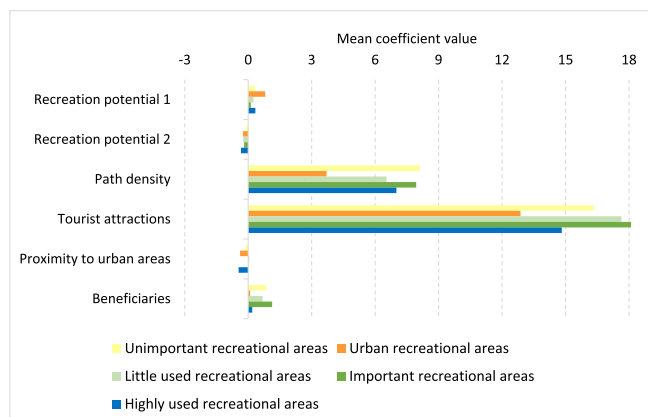


Fig. A2. Mean estimated regression coefficients of the variables included in the GWR model for the five clusters.

Table A1

Moran's *I* and related co-variables for testing spatial autocorrelation for recreation. Moran's *I* values close to +1 indicate high spatial clustering of high values and/or low values.

	Moran's index	z-Score	p-Value
Supply	0.69	342.50	<0.0001
Demand	0.38	187.45	<0.0001
Flow	0.56	275.86	<0.0001
Flow spring	0.40	197.23	<0.0001
Flow summer	0.44	215.04	<0.0001
Flow autumn	0.42	207.29	<0.0001
Flow winter	0.38	188.54	<0.0001

Appendix B. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.ecoser.2017.11.017>.

Table A2

Results of the visual content analysis of Flickr photographs.

Municipality	Country	Area (km ²)	Topography [*]	Number of pictures	Share of pictures related to recreation	Number of photos per user
Isenthal	CH	61	1	84	83.33	2.55
Sisikon	CH	17	1	176	82.95	1.96
Glarus Nord	CH	147	1	394	90.61	2.86
Oberdorf (SO)	CH	12	1	173	89.60	3.53
Bonaduz	CH	14	1	179	88.83	2.49
Ftan	CH	43	1	151	82.12	3.60
Waltensburg/Vuorz	CH	32	1	59	69.49	3.69
Centovalli	CH	51	1	360	61.39	4.19
Frasco	CH	26	1	78	92.31	2.89
Ried-Brig	CH	48	1	169	87.57	3.02
Fiesch	CH	11	1	192	97.40	2.26
La Chaux-de-Fonds	CH	56	1	206	52.91	4.12
Vernier	CH	8	2	60	55.00	2.22
Livet-et-Gavet	FR	62	1	114	88.60	3.00
Pierrevert	FR	28	1	149	55.03	12.42
La Chapelle-en-Valgaudemar	FR	128	1	232	93.53	5.27
Cap-d'Ail	FR	2	1	106	64.15	1.39
Mazaugues	FR	54	1	84	90.48	5.60
Monieux	FR	48	1	218	94.95	4.11
Acceglio	IT	151	1	677	84.19	9.03
Chiusa di Pesio	IT	95	1	125	88.00	2.66
Tortona	IT	99	2	98	38.78	2.13
La Salle	IT	84	1	111	83.78	1.95
Valdidentro	IT	226	1	491	92.26	2.77
Braone	IT	13	1	62	82.26	6.89
Rasun-Anterselva	IT	122	1	678	88.50	4.88
Grigno	IT	46	1	75	93.33	1.63
Mezzocorona	IT	25	1	84	38.10	4.20
Legnago	IT	79	2	136	30.15	9.07
Forni Avoltri	IT	81	1	194	92.78	6.47
Offenburg	DE	78	1	122	68.85	2.39
Bisingen	DE	33	1	397	48.61	2.12
Ettal	DE	15	1	998	94.79	2.55
Kreuth	DE	122	1	543	96.32	5.48
Leipheim	DE	32	2	60	28.33	2.61
Wertach	DE	46	1	65	60.00	2.71
Gaming	AT	244	1	139	94.24	5.56
Tullnerbach	AT	20	2	54	62.96	1.86
Zwettl-Niederösterreich	AT	256	1	176	51.70	2.79
Mühlbach am Hochkönig	AT	52	1	240	79.58	4.36
Sankt Margarethen im Lungau	AT	24	1	99	89.90	4.50
Söll	AT	46	1	215	90.70	3.41
Serfaus	AT	60	1	276	95.29	4.25
Brda	SI	72	1	147	55.78	2.63
Maribor	SI	148	1	472	71.82	4.54
Metlika	SI	109	1	70	31.43	4.12
Radovljica	SI	119	1	104	51.92	1.58
Total				10092	80.30	3.77

^{*} 1 = mountainous, 2 = lowland.

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